

Chapter 4:

Advanced Derivative Applications

Chapter 4 Overview:

There are two main contexts for derivatives: graphing and motion. In this chapter, we will consider both these applications of the derivative. Much of this is a review of material covered in precalculus. Key topics include:

- Extremes
- The First and Second Derivative Tests
- Optimization
- Derivative-related theorems
- Intervals of increasing and decreasing
- Advanced graphing using the derivative
- Rectilinear motion

This chapter makes up a solid portion of the multiple choice, along with also being at least one full free response question (and often parts of others) on the AP exam.

4.1: Extrema and the Derivative Tests

One of the most valuable aspects of calculus is that it allows us to find extreme values of functions. The ability to find maximum or minimum values of functions has wide-ranging applications. Every industry has uses for finding extremes, from optimizing profit and loss, to maximizing output of a chemical reaction, to minimizing surface areas of packages. This one tool of calculus eventually revolutionized the way the entire world approached every aspect of industry. It allowed people to solve formerly unsolvable problems.

OBJECTIVES

Find Critical Values and Extreme Values for Functions

Use the 1st and 2nd Derivative Tests to Identify Maxima v.s. Minima

But, before we begin, let's review some definitions from last year, which will be useful in this chapter as well as subsequent chapters.

Critical Value → The x -coordinate of the extreme.

Maximum Value → The y -coordinate of the high point.

Minimum Value → The y -coordinate of the low point.

Relative Extremes → The highest or lowest point in a given section of the curve.

Absolute Extremes → The highest or lowest point of the whole curve.

Interval of Increasing → The interval of x -values for which the curve is rising from left to right.

Interval of Decreasing → The interval of x -values for which the curve is dropping from left to right.

Critical values of a function occur when:

1) $\frac{dy}{dx} = 0$

2) $\frac{dy}{dx}$ does not exist

3) the function reaches an endpoint of its domain

It's also helpful to remember that a **critical value** is referring specifically to the value of the x , while the **extreme value** refers to the value of the y .

As we already know, the first derivative is what allows us to algebraically find extremes, and

the First Derivative Test allows us to interpret critical values as maxima or minima. Since the first derivative of a function tells us when that function is increasing or decreasing, we can figure out if a critical value is associated with a maximum or minimum depending on the sign change of the derivative. Let's quickly refresh our memory on the First Derivative Test.

The First Derivative Test

As the sign pattern of the first derivative is viewed left to right, the critical value represents a:

- 1) relative maximum if the sign changes from $+$ to $-$
- 2) relative minimum if the sign changes from $-$ to $+$
- 3) neither a relative max or min if the sign does not change

Ex 4.1.1: Apply the First Derivative Test to the function $y = 5x^4 - 10x^2$.

Sol 4.1.1:

$$\frac{dy}{dx} = 20x^3 - 20x = 0$$

$$20x(x^2 - 1) = 0$$

$$x = -1, 0, 1$$

Now, let's make our sign pattern.

$$\begin{array}{ccccccc} y' & - & 0 & + & 0 & - & 0 & + \\ & \leftarrow & & & & & & \rightarrow \\ x & & -1 & & 0 & & 1 & \end{array}$$

So, there appear to be critical values at x values of -1 , 0 , and 1 . By utilizing the First Derivative Test, we can see that we have a minimum at -1 , a maximum at 0 , and another minimum at 1 .

Ex 4.1.2: Find the maximum values for the function $g(x) = \sqrt{x^3 - 9x}$.

Sol 4.1.2: For this question, we have to be careful! Because we are dealing with a square root, we will have domain restrictions. We can find such restrictions by starting with a sign pattern for $g(x)$.

$$\begin{array}{ccccccc}
 g(x) & - & 0 & + & 0 & - & 0 & + \\
 & \longleftarrow & & & & & & \longrightarrow \\
 x & & -3 & & 0 & & 3 &
 \end{array}$$

Now, since we know that the argument of a square root has to be positive, we know that our domain is when $x^3 - 9x \geq 0$. Therefore, based on our sign pattern, our domain is $x \in [-3, 0] \cup [3, \infty)$.

With that out the way, we can now take the first derivative and apply the First Derivative Test.

$$g'(x) = \frac{3x^2 - 9}{2\sqrt{x^3 - 9x}}$$

$$g'(x) = 0 \text{ when } 3x^2 - 9 = 0 \therefore x = \pm\sqrt{3}$$

$$g'(x) = DNE \text{ when } x^3 - 9x = 0 \therefore x = \pm 3, 0$$

Since $x = \sqrt{3}$ is not in the domain of the function, our critical values are at $x = \pm 3, 0, -\sqrt{3}$. Therefore, our sign pattern looks like:

$$\begin{array}{ccccccc}
 g'(x) & DNE & + & 0 & - & DNE & & DNE & + \\
 & \longleftarrow & & & & & & & \longrightarrow \\
 x & & -3 & & -\sqrt{3} & & 0 & & \sqrt{3} & & 3
 \end{array}$$

From our sign pattern, we can conclude that at $x = -\sqrt{3}$, we have a maximum value, and by substituting this value into the function, we find the maximum value is at $y = 3.224$.

The First Derivative Test is not the only way to test whether critical values are associated with maximum or minimum values. If you recall, the second derivative can show us intervals of concavity. This is very useful because if we know that the curve is concave down at a critical value, it must be associated with a maximum. Similarly, if the curve is concave up, the critical value must be associated with a minimum value.

The Second Derivative Test

For a function f :

- 1) If $f'(c) = 0$ and $f''(c) > 0$, then f has a relative minimum at c
- 2) If $f'(c) = 0$ and $f''(c) < 0$, then f has a relative maximum at c

Now, we can revisit **Ex 1.5.7** part (c), which we previously could not do without the Second Derivative Test.

Ex 4.1.3: Consider the curve given by $x^2 + 4xy + y^2 = -12$.

- (c) Find the value(s) of $\frac{d^2y}{dx^2}$ at the point(s) found in part (b). Does the curve have a local maximum, a local minimum, or neither at those points? Justify your answer.

(Ex 1.5.7)

Sol 4.1.3: What is really asked here is to apply the Second Derivative Test, because we cannot create a sign pattern for non-functions. The y is not isolated in the equation. Recall that the two points we found in part (b) were $(-4, 2)$ and $(4, -2)$.

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx} \left[\frac{dy}{dx} = -\frac{x+2y}{2x+y} \right] \\ &= \frac{(2x+y) \left(1 + 2\frac{dy}{dx} \right) - (x+2y) \left(2 + \frac{dy}{dx} \right)}{(2x+y)^2} \\ \frac{d^2y}{dx^2} \Big|_{(-4,2)} &= \frac{(2(-4)+2)(1+2(0)) - (-4+2(2))(2+0)}{(2(-4)+2)^2} \\ &= \frac{6}{(-6)^2} > 0 \\ \frac{d^2y}{dx^2} \Big|_{(4,-2)} &= \frac{(2(4)-2)(1+2(0)) - (4+2(-2))(2+0)}{(2(4)+2)^2} \\ &= \frac{6}{(-6)^2} < 0\end{aligned}$$

$(-4, 2)$ will be a minimum because the second derivative is positive.

$(4, -2)$ will be a maximum because the second derivative is negative.

Ex 4.1.4: Use the Second Derivative Test to determine if the critical values of $g(t) = 27t - t^3$ are at a maximum or minimum value.

Sol 4.1.4:

$$g'(t) = 27 - 3t^2 = 0$$

$$t = \pm 3$$

So, -3 and 3 are critical values. Now, let's take a look at the second derivative of $g(t)$ at those values.

$$g''(t) = -6t$$

$$g''(-3) = -6(-3) = 18$$

$$g''(3) = -6(3) = -18$$

Finally, we can apply the Second Derivative Test. Since $g'(-3) = 0$ and $g''(-3) > 0$, we can determine that $t = -3$ is a minimum. Similarly, since $g'(3) = 0$ and $g''(3) < 0$, we can determine that $t = 3$ is a maximum. Note that the numerical value of the second derivative is irrelevant — only the sign matters.

Ex 4.1.5: Determine if the critical values of $y = (3 - x^2)e^x$ are at a maximum or a minimum value.

Sol 4.1.5: First, let's find the critical values.

$$\frac{dy}{dx} = (3 - x^2)e^x + e^x(-2x) = 0$$

$$-e^x(x^2 + 2x - 3) = 0$$

$$-e^x(x + 3)(x - 1) = 0$$

From here, we can see that our critical values are $x = -3, 1$. The derivative is never *DNE*. So, let's find the second derivative and substitute in our values.

$$\frac{dy}{dx} = -e^x(x^2 + 2x - 3)$$

$$\begin{aligned}\frac{d^2y}{dx^2} &= -e^x(2x + 2) + (x^2 + 2x - 3)(-e^x) \\ &= -e^x(x^2 + 4x - 1)\end{aligned}$$

$$\left. \frac{d^2 y}{dx^2} \right|_{x=1} = -4e$$

$$\left. \frac{d^2 y}{dx^2} \right|_{x=-3} = 4e^{-3}$$

By the Second Derivative Test, we can determine that $x = 1$ is a maximum and $x = -3$ is a minimum.

Optimization

Optimization is a practical application of finding maxima and minima for functions. As stated before, this idea revolutionized thinking and is a critical component of all industries. We've previously approached this topic in precalculus, but now, this concept forms a much more fundamental part of what we need to be able to do.

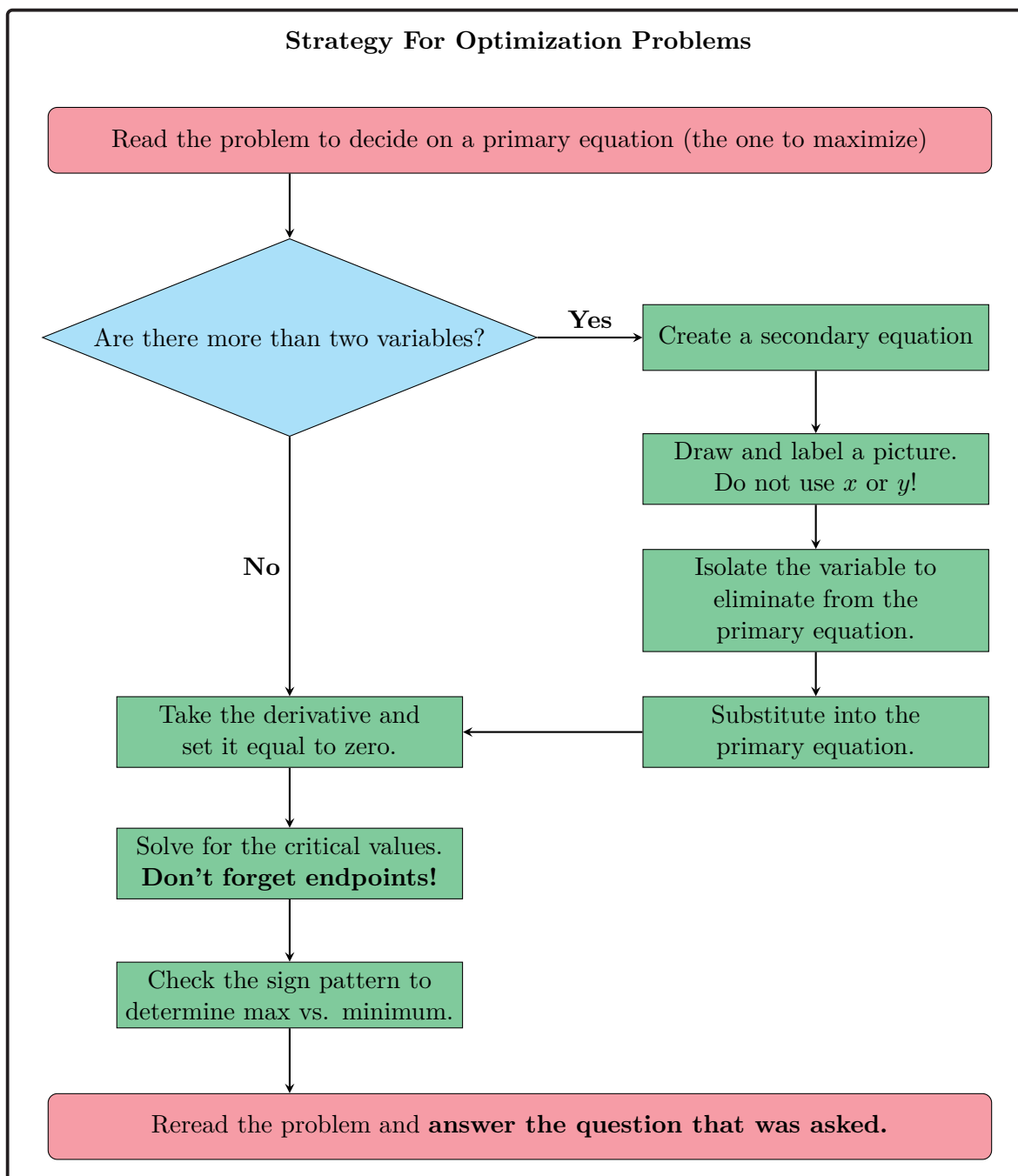
OBJECTIVES

Solve Optimization Problems.

Every optimization problem looks a bit different, but they all follow a similar progression. You must first identify your variables and any formula you need. Use algebra to eliminate any variables, and take the derivative of **the function you're trying to optimize**. That is the most common mistake in optimization: taking the derivative of the wrong function. Now, let's take a look at the following example:

Ex 4.1.6: The volume of a cylindrical cola can is 32π in³. What is the minimum surface area for such a can?

The word “minimum” tells us that we have an optimization problem. Recall our workflow for tackling optimization problems:



So, let's tackle our example.

Sol 4.1.6: The problem asks to minimize surface area, which is determined by:

$$S = 2\pi r^2 + 2\pi rh$$

As there are more than two variables in this equation, either r or h needs to be elimi-

nated in this formula before differentiating. The volume is $V = \pi r^2 h = 32\pi$, so $h = \frac{32}{r^2}$ and

$$S = 2\pi r^2 + 2\pi r \left(\frac{32}{r^2} \right)$$

$$= 2\pi r^2 + \frac{64\pi}{r}$$

$$S' = 4\pi r - \frac{64\pi}{r^2} = 0$$

$$\therefore 4\pi r = \frac{64\pi}{r^2}$$

$$r^3 = 16 \rightarrow r = 2.5198$$

Now, let's make a sign pattern to determine if this critical value is a minimum.

$$\begin{array}{ccccccc} S' & & - & & 0 & & + \\ & \longleftarrow & & & & & \longrightarrow \\ r & & & & 2.520 & & \end{array}$$

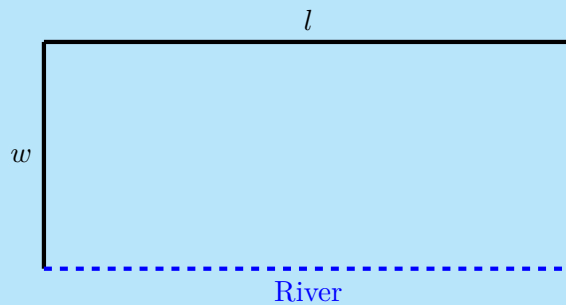
Because the derivative changes from negative to positive, we know that 2.5198 is a minimum. We can plug it back into our surface area equation to find our minimum surface area.

$$S(2.5198) = 2\pi(2.5198)^2 + \frac{64\pi}{2.5198}$$

$$S(2.5198) = 119.687 \text{ in}^2$$

Therefore, the minimum surface area of the cola can is $\boxed{119.687 \text{ in}^2}$.

Ex 4.1.7: The owner of the Rancho Grande has 3000 yards of fencing material with which to enclose a rectangular piece of grazing land along the straight portion of a river. If fencing is not required along the river, what are the dimensions of the largest area he can enclose? What is the area?



Sol 4.1.7: We first begin with our area formula, $A = lw$. Now, as we know, we need to express one variable in terms of the other. We can do this by utilizing the fact about perimeter to complete the problem.

$$P = l + 2w$$

$$3000 = l + 2w$$

$$3000 - 2w = l$$

Let's substitute this back into our area formula.

$$A = (3000 - 2w)w$$

$$A = 3000w - 2w^2$$

Now, since we have an equation with one independent variable, we can take the derivative easily.

$$\frac{dA}{dw} = 3000 - 4w = 0$$

$$w = 750$$

$$l = 1500$$

So, we would want a width of 750 yards and a length of 1500 yards. This would give us an area of 1,125,000 yd².

Ex 4.1.8: Find the point on the curve $y = \frac{e^{-x^2}}{2}$ that is closest to the origin.

Sol 4.1.8: We want to minimize the distance to the origin, so we will be using the Pythagorean Theorem to find the distance.

$$D = \sqrt{x^2 + y^2}$$

$$D = \sqrt{x^2 + \left(\frac{e^{-x^2}}{2}\right)^2}$$

$$\frac{dD}{dx} = \frac{1}{2} \left(x^2 + \frac{e^{-2x^2}}{4}\right)^{-\frac{1}{2}} (2x - xe^{-2x^2}) = 0$$

$$x = 0, \pm 0.841$$

$\frac{dD}{dx}$	-	0	+	0	-	0	+
	←—————→						
x		-0.841		0		0.841	

By the First Derivative Test, we know that $x = 0$ is a minimum of the curve. Plugging that value back into our original equation gives us our desired point of $\boxed{(0, 0.5)}$.

4.1 Free Response Homework

Find the critical and extreme values for each function.

1. $y = 2x^3 + 9x^2 - 168x$

2. $y = 3x^4 + 2x^3 + 12x^2 + 12x - 42$

3. $y = \frac{x^2 + 1}{x^3 - 4x}$

4. $y = \frac{x - 5}{x^2 + 9}$

5. $y = \sqrt{9x^3 - 4x^2 - 27x + 12}$

6. $y = \sqrt{\frac{3x}{9 - x^2}}$

7. $y = x^2 \sqrt[3]{9 - x^2}$

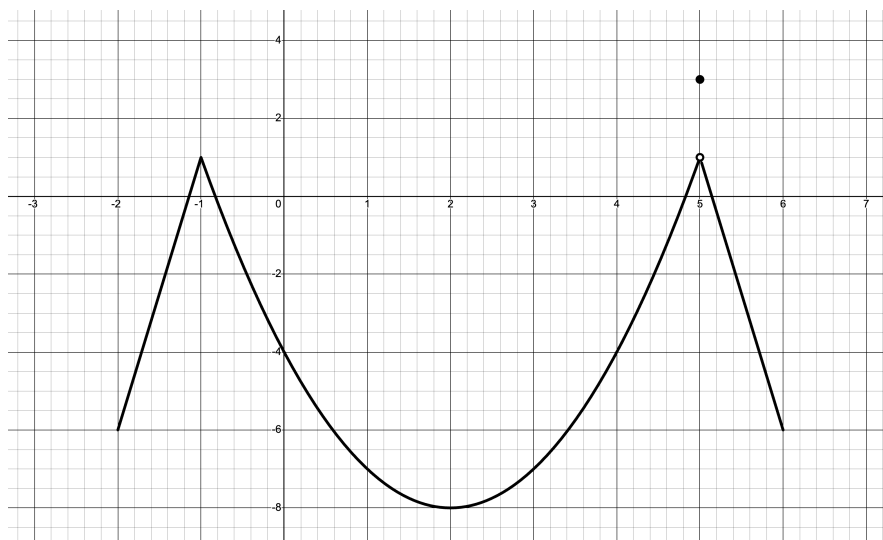
8. $y = (x - x^2) 3^x$

9. $y = \frac{\sqrt{3}}{2}x + \cos(x) \quad \left| \quad x \in [-2\pi, 2\pi] \right.$

10. $y = \cot^{-1}\left(\frac{1}{x}\right) - \tan^{-1}(x)$

11. $y = \ln(x^2 + 4x + 14)$

12.



For 13 – 18, find the absolute maximum and minimum values of f on the given intervals.

13. $f(t) = \frac{x}{x^2 + 1} \quad \left| \quad x \in [0, 2] \right.$

14. $f(t) = \sqrt[3]{t}(8 - t) \quad \left| \quad t \in [0, 8] \right.$

15. $f(z) = ze^{-z} \quad \left| \quad z \in [0, 2] \right.$

16. $f(x) = \frac{\ln x}{x} \quad \left| \quad x \in [1, 3] \right.$

$$17. f(x) = e^{-x} - e^{-2x} \quad \left| \quad x \in [0, 1] \right.$$

$$18. f(y) = \sin\left(\frac{y}{3}\right) + \frac{2y}{9} \quad \left| \quad y \in [-10, 15] \right.$$

For each of the following functions, apply the First Derivative Test, then verify the results with the Second Derivative Test.

$$19. f(x) = x^2 \ln x$$

$$20. h(f) = f^3 - 12f + 21$$

$$21. f(x) = xe^{-x^2}$$

$$22. h(t) = e^{t^3+t^2+1}$$

Solve these problems algebraically.

23. Find two positive numbers whose product is 110 and whose sum is a minimum.

24. Find a positive number such that the sum of the number and its reciprocal is a minimum.

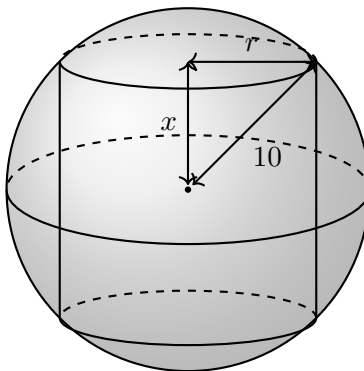
25. A farmer with 750 feet of fencing material wants to enclose a rectangular area and divide it into four smaller rectangular pens with sides parallel to one side of the rectangle. What is the largest possible total area.

26. If 1200 cm^2 of material is available to make a box with an open top and a square base, find a maximum volume the box can contain.

28. Find the point(s) on the curve $y = \frac{1}{x^2 + 1}$ that is/are closest to the origin.

29. Find the area of the largest rectangle that can be inscribed in the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

30. Find the maximum volume of a cylinder inscribed in a sphere with radius 10.



31. A piece of wire 10 m long is cut into two pieces. One piece is bent into a square, while the other is bent into an equilateral triangle.

(a) Find where the wire should be cut to maximize the area enclosed.

(b) Find where the wire should be cut to minimize the area enclosed.

32. You need to enclose 500 cm^3 of fluid in a cylinder. If the material you are using costs 0.001 dollars per cm^3 , find the size of the cylinder that minimizes the cost. If the product that you are containing is a sports drink, do you think that the size that minimizes the cost is the most efficient size? Explain

33. The height of a man jumping off of a high dive is given by the function $h(t) = -4.9t^2 + 2t + 10$ on the domain $0 \leq t \leq 1.64715$. Find the absolute maximum and minimum heights modeled by this function.

34. Given the area of a triangle can be calculated with the formula $A = \frac{1}{2}ab\sin(\theta)$, what value of θ will maximize the area of a triangle (given that a and b are constants)?

35. You operate a tour service that offers the following rates for the tours: \$200 per person if the minimum number of people book the tour (50 people is the minimum) and for each person past 50, up to a maximum of 80 people, the cost per person is decreased by \$2. It costs you \$6000 to operate the tour plus \$32 per person.

(a) Write a function that represents cost, $C(x)$.

(b) Write a function that represents revenue, $R(x)$.

(c) Given that profit can be represented as $P(x) = R(x) - C(x)$, write a function that represents profit and state the domain for the function.

(d) Find the number of people that maximizes your profit. What is the maximum profit?

4.1 Multiple Choice Homework

1. Give the value of x where the function $f(x) = x^3 - 9x^2 + 24x + 4$ has a relative maximum point.

a) 4

b) -2

c) 2

d) -4

e) 3

2. Give the value of x where the function $f(x) = x^3 - \frac{33}{2}x^2 + 84x - 2$ has a relative minimum point.

- a) -4 b) -7 c) 4 d) 5 e) 7
-

3. Give the approximate location of a relative maximum point for the function $f(x) = 3x^3 + 5x^2 - 3x$.

- a) $(-1.357, 5.779)$ b) $(0.2457, -0.3908)$ c) $(-1.357, 5.713)$
d) $(0.2457, -0.3216)$ e) $(-1.357, -0.3908)$
-

4. Consider the following sign pattern:

$$\begin{array}{ccccccccccc} \frac{dy}{dx} & & + & 0 & & 0 & - & 0 & + & 0 & & 0 & - \\ \leftarrow & & & & & & & & & & & & \rightarrow \\ x & & & -2 & & -\sqrt{2} & & 0 & & \sqrt{2} & & 2 & \end{array}$$

What value of x does y has a local minimum?

- a) -2 b) $-\sqrt{2}$ c) 0 d) $\sqrt{2}$ e) 2
-

5. Find the absolute maximum of $y = -\sqrt{25 - x^2}$ on the interval $x \in [-2, 4]$.

- a) -2 b) 0 c) -5 d) $-\sqrt{21}$ e) -3
-

6. Find the absolute maximum value of $y = \sqrt{36 - x^2}$ on the interval $x \in [-2, 2]$.

- a) 5 b) 6 c) 7 d) 0 e) 1
-

7. The graph of the function $f(x) = 2x^{\frac{5}{3}} - 5x^{\frac{2}{3}}$ is increasing on which of the following intervals.

- I. $1 < x$ II. $0 < x < 1$ III. $x < 0$

- a) I only b) II only c) III only d) I and II only e) II and III only

8. If $f(x) = x^2e^{-2x}$, then the graph of f is increasing for all x such that

- a) $0 < x < 1$ b) $0 < x < \frac{1}{2}$ c) $0 < x < 2$ d) $x < 0$ e) $x > 0$
-

9. If $f(x) = \frac{x}{\ln(7x)}$, then what is the interval of decreasing?

- a) $\left(0, \frac{1}{7}\right)$ b) $\left(0, \frac{1}{7}\right) \cup \left(\frac{1}{7}, \frac{1}{7}e\right)$ c) $(1, 7e)$ d) $(1, 7)$ e) $\left(1, \frac{1}{7}e\right)$
-

10. Suppose $f'(x) = \frac{(x+1)^3(x-4)^2}{(x^4+1)}$. Which of the following statements must be true?

I. The slope of the line tangent to $y = f(x)$ at $x = 1$ is 36.

II. $f(x)$ is increasing on $x \in (1, 4)$.

III. $f(x)$ has a minimum at $x = -1$.

- a) II only b) I and III only c) II and III only d) I and II only e) I, II, and III
-

11. Consider the following sign pattern:

$$\begin{array}{ccccccc} f'(x) & - & 0 & + & 0 & - & 0 & - \\ & \longleftarrow & & & & & & \longrightarrow \\ x & & -4 & & -1 & & 2 & \end{array}$$

- a) -4 b) -1 c) 1 d) 2 e) None of these
-

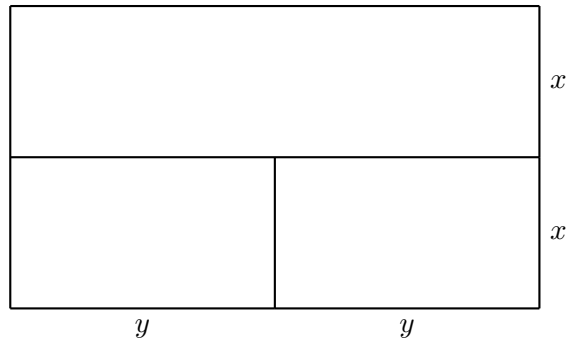
12. The sum of two positive integers is x and y is 150. Find the value of x that minimizes $P = x^3 - 150xy$.

- a) $x = 25$ b) $x = 50$ c) $x = 75$ d) $x = 100$ e) $x = 125$
-

13. The sum of two positive integers x and y is 120. Find the value of x that minimizes $P = x^3 - 120xy$.

a) $x = 20$ b) $x = 40$ c) $x = 60$ d) $x = 80$ e) $x = 100$

14. A farmer has 100 yards of fence to enclose a field, subdivide it into two equal pens, and further subdivide one of those pens into two equal fields as shown below.

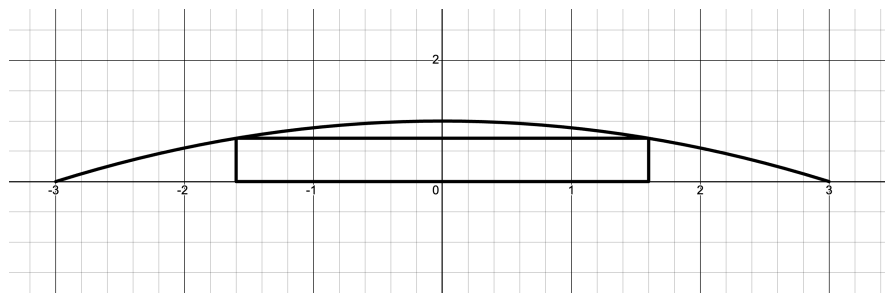


a) 12.5 b) 10 c) $\frac{100}{11}$ d) $\frac{25}{3}$ e) None of these

15. A farmer with 890 ft of fencing wants to enclose a rectangular area and then divide it into four pens with fencing parallel to one side of the rectangle. What is the largest possible total area of the four pens?

a) 19,825.5 ft² b) 19,805.5 ft² c) 19,801.5 ft² d) 19,902.5 ft² e) 19,791.5 ft²

16. A rectangle with one side on the x -axis has its upper vertices on the graph of $y = 1 - \frac{x^2}{9}$, as shown in the figure below. What is the maximum area of the rectangle?



- a) $\sqrt{3}$ b) $\frac{\sqrt{3}}{2}$ c) $\frac{2}{3}$ d) $\frac{4\sqrt{3}}{3}$ e) None of these
-

17. In a certain community, an epidemic spreads in a way that the percentage P of the population that is infected after t months is modeled by

$$P(t) = \frac{kt^2}{(C + t^2)^2},$$

where C and k are constants. Find t , such that P is maximized.

- a) 0 b) $\sqrt{C}$ c) $\sqrt{k}$ d) $\sqrt{\frac{C}{3}}$ e) None of these
-

18. Consider the following table below:

x	$g(x)$	$g'(x)$	$g''(x)$	$f(x)$	$f'(x)$	$f''(x)$
0	2	3	-2	2	1	-2
2	4	2	0	4	0	3
4	1	-1	3	0	2	-1

At $x = 2$, $f(x)$ has a:

- a) Local min. b) Local max. c) Point of inflection d) Zero e) None of these
-

19. Let $f(x)$ and $g(x)$ be twice-differentiable functions with selected values given in the table in **question 18**. Let $h(x) = g(f(x))$. At $x = 2$, $h(x)$ has a:

- a) Local min. b) Local max. c) Point of inflection d) Zero e) None of these
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4.2: The Mean Value and Rolle's Theorem

The Mean Value Theorem is an interesting piece of the history of calculus that was used to prove a lot of what we take for granted. The Mean Value Theorem was used to prove that a derivative being positive or negative told you that the function was increasing or decreasing, respectively. Of course, this led directly to the First Derivative Test and the intervals of concavity.

Mean Value Theorem

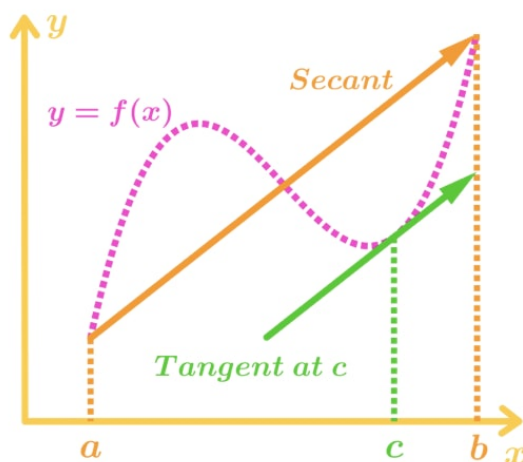
If f is a function that satisfies these two hypotheses:

- 1) f is continuous on the closed interval $[a, b]$.
- 2) f is differentiable on the closed interval (a, b) .

Then there exists a number c in the interval (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Again, translating from math to english, this simply states that if you have a smooth, continuous curve, the slope of the line connecting the endpoints has to equal the slope of a tangent somewhere in that interval. Alternatively, it says that the secant line (a line through two distinct points on a curve) through the endpoints will have the same slope as a tangent line at some point on the curve.



Ex 4.2.1: Show that the function $f(x) = x^3 - 4x^2 + 1$ with $x \in [-1, 3]$ satisfies all the conditions of the Mean Value Theorem and find c .

Sol 4.2.1: Polynomials are continuous throughout their domain, so the first hypothesis is satisfied. Polynomials are also differentiable throughout their domain, so the second condition is satisfied.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = 3c^2 - 8c = \frac{f(3) - f(-1)}{3 - (-1)}$$

$$3c^2 - 8c = \frac{-8 - (-4)}{4}$$

$$3c^2 - 8c = -1 \implies 3c^2 - 8c + 1 = 0$$

$$c = \boxed{2.535, 0.131}$$

For this problem, we actually find *two* values of c . This is definitely possible, as the Mean Value Theorem doesn't state how many numbers c that satisfy the theorem; simply that there exists at least one number c .

Ex 4.2.2: Show that the function $f(x) = x^2 - 4x + 1$ with $x \in [0, 4]$ satisfies all the conditions of the Mean Value Theorem and find c .

Sol 4.2.2: Polynomials are continuous throughout their domain, so the first hypothesis is satisfied. Polynomials are also differentiable throughout their domain, so the second condition is satisfied.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = 2c - 4 = \frac{f(4) - f(0)}{4 - 0}$$

$$2c - 4 = \frac{1 - 1}{4}$$

$$2c - 4 = 0$$

$$c = \boxed{2}$$

Now, let's talk a little about Rolle's Theorem. Rolle's Theorem is a specific case of the Mean Value Theorem (though Joseph-Louis Lagrange used it to prove the Mean Value Theorem).

Rolle's Theorem is extremely powerful, as it was used to prove all of the rules we have used to interpret derivatives. It used to be a very useful theorem, but is now something of a historical curiosity.

Rolle's Theorem

If f is a function that satisfies these three hypotheses:

- 1) f is continuous on the closed interval $[a, b]$.
- 2) f is differentiable on the closed interval (a, b) .
- 3) $f(a) = f(b)$

Then there exists a number c in the interval (a, b) such that $f'(c) = 0$.

Although this theorem may seem confusing at first, it is simply stating that if you have a continuous and smooth curve with an initial point and ending point at the same height, then there exists some point in the curve that has a derivative of zero. This can be simply proved by utilizing the Mean Value Theorem and noticing that $f(b) - f(a) = 0$.

$$f'(c) = \frac{f(b) - f(a)}{b - a} \rightarrow f'(c) = \frac{0}{b - a} \implies f'(c) = 0$$

Another theorem that is important to note and often confused with Rolle's Theorem or the Mean Value Theorem is the **Extreme Value Theorem**. Recall what the Extreme Value Theorem states:

Extreme Value Theorem

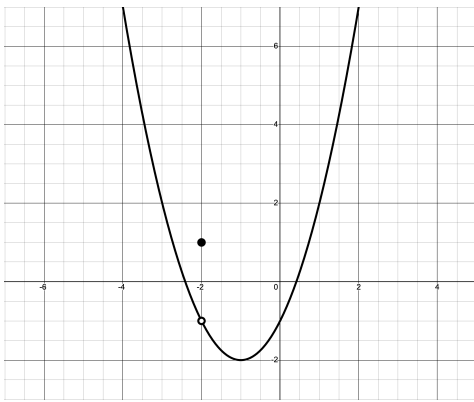
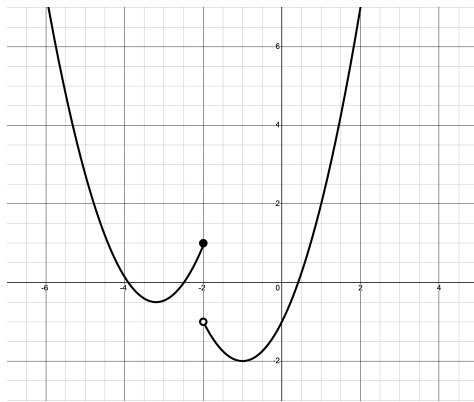
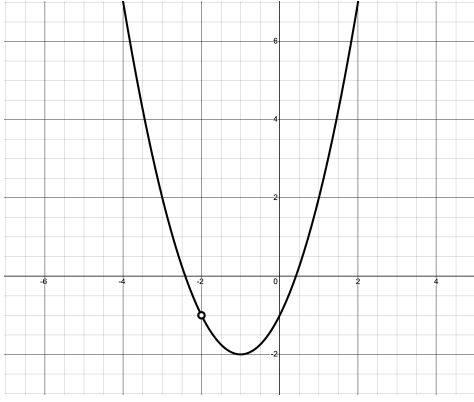
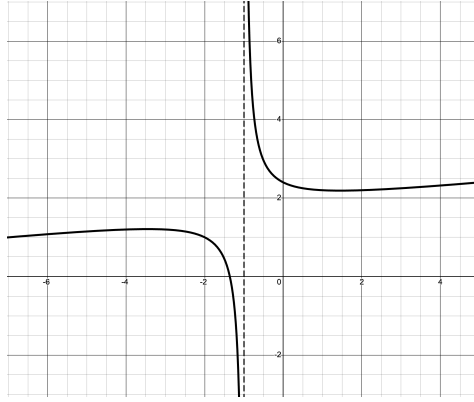
If f is continuous on the closed interval $[a, b]$, then $f(x)$ attains a maximum and a minimum at least once each.

The final theorem that may be confused with the Mean Value Theorem is the **Intermediate Value Theorem**. Recall what the Intermediate Value Theorem states:

Intermediate Value Theorem

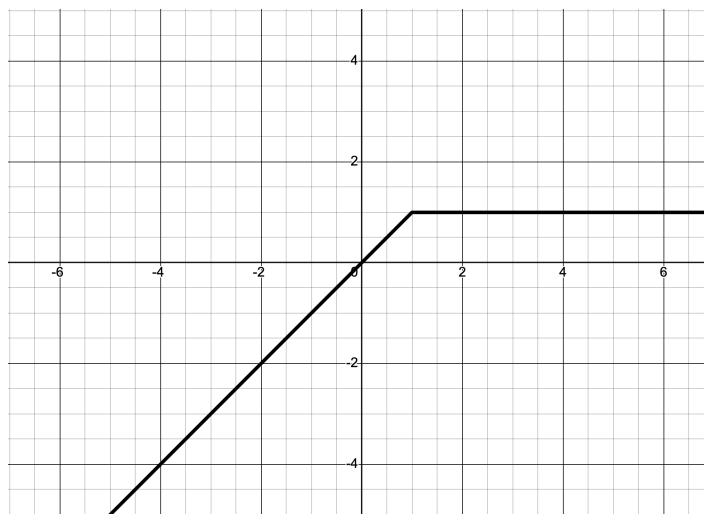
If f is continuous on the closed interval $[a, b]$, then $f(x)$ attains every height (or value) between $f(a)$ and $f(b)$.

Now that we've gone over the Mean Value Theorem, we need to consider when this theorem might not apply to a problem — that is, when is a function not continuous or not differentiable. We first saw the concept of non-continuous functions in precalculus, but let's formalize these discontinuities within the following table.

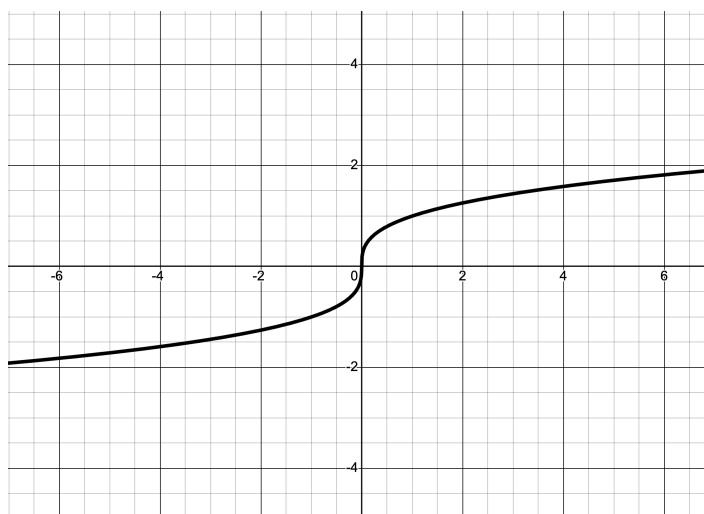
	Removable Discontinuity $\left(\lim_{x \rightarrow a} f(x) \text{ Exists } \right)$	Essential Discontinuity $\left(\lim_{x \rightarrow a} f(x) \text{ Does Not Exist } \right)$
$f(a)$ Exists	 <i>Point of Displacement (POD)</i>	 <i>Jump Discontinuity</i>
$f(a)$ Does Not Exist	 <i>Point of Exclusion (POE)</i>	 <i>Vertical Asymptote</i>

Moving on, let's talk about functions that are not differentiable. "Not differentiable" simply means the derivative does not exist at some point in the function. This can happen one of three ways:

1. A function is not differentiable if it is not continuous (see table above).
2. A function is not differentiable if at some point, the tangent line at that point is vertical.



3. A function is not differentiable if it is not "smooth".



The Mean Value Theorem often arises in AP Table Problems like this one:

Ex 4.2.3: Below is a chart showing specific values of the rate of sewage flowing through a pipeline according to time in minutes.

t (minutes)	0	4	6	10	13	15	20
$V(t)$ (gallons/min)	83	68	83	48	38	30	38

Assume $V(t)$ is a continuous and differentiable function. Explain why there are at least two times between $t = 0$ and $t = 20$ when $V'(t) = 0$.

Sol 4.2.3: Because $V(t)$ is a continuous and differentiable function, the Mean Value Theorem applies. Since $V(0) = V(6) = 83$, we have

$$V'(t) = \frac{V(6) - V(0)}{6 - 0} = \frac{0}{6} = 0 \quad \checkmark$$

Similarly, since $V(13) = V(26) = 38$, we have

$$V'(t) = \frac{V(26) - V(13)}{26 - 13} = \frac{0}{13} = 0 \quad \checkmark$$

There may be more c -values where $V'(t) = 0$, but as we can see, we have at least two.

Astute students may notice that we can also apply Rolle's Theorem to this problem, since $f(a) = f(b)$!

Ex 4.2.4: Below is a chart showing specific values of the rate of sewage flowing through a pipeline according to time in minutes.

t (minutes)	0	4	6	10	13	15	20
$V(t)$ (gallons/min)	83	68	83	48	38	30	38

Assume $V(t)$ is a continuous and differentiable function. Explain why there are at least two times between $t = 0$ and $t = 20$ when $V'(t) = 74$.

Sol 4.2.4: For this problem, we have to be a little careful! The Mean Value Theorem would not be very useful in this problem, as we cannot find a $\frac{V(b) - V(a)}{b - a}$ that is equal to 74. Instead, let's turn to another theorem in our arsenal: the Intermediate Value Theorem.

Because $V(t)$ is continuous and $68 < 74 < 83$, according to the Intermediate Value Theorem, $V(t)$ will attain the height (or value) of 74 between $V(0) = 83$ and $V(4) = 68$, and $V(4) = 68$ and $V(6) = 83$. Thus, we have found our two times where $V(t) = 74$.

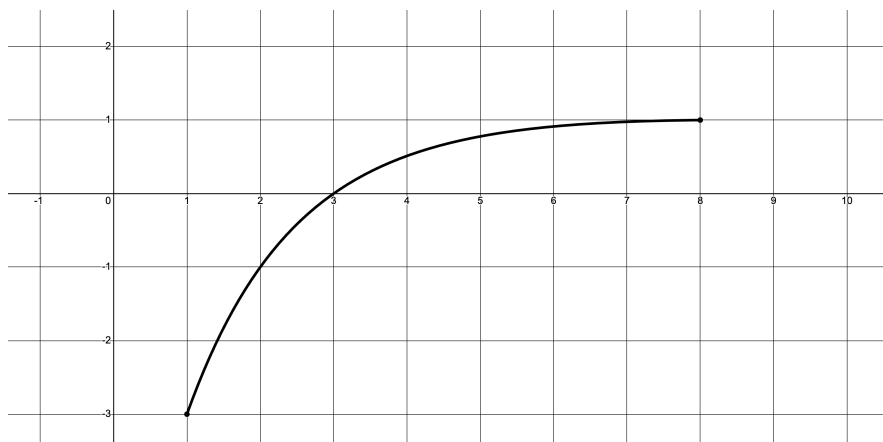
4.2 Free Response Homework

Verify that the following functions fit all the conditions of the Mean Value Theorem, and then find all values of c that satisfy the conclusion of the Mean Value Theorem.

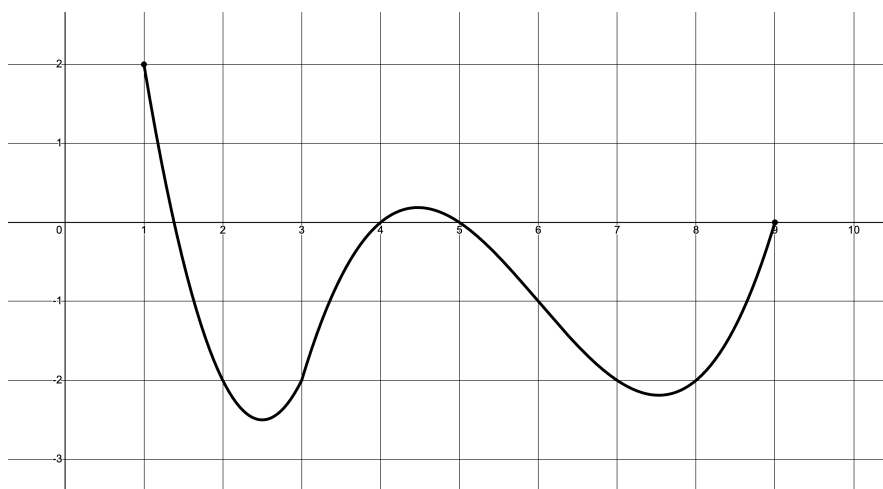
1. $f(x) = x^3 - 3x^2 + 2x + 5$ $x \in [0, 2]$ 2. $f(x) = \sin(2\pi x)$ $x \in [-1, 1]$

3. $g(t) = t\sqrt{t+6}$ $t \in [-4, 0]$ 4. $H(t) = \frac{t}{(t-6)^2}$ $t \in [2, 8]$

5. Given the graph of the function below, estimate all values of c that satisfy the conclusion of the Mean Value Theorem for the interval $[1, 8]$.



6. of the function below, estimate all values of c that satisfy the conclusion of the Mean Value Theorem for the interval $[1, 9]$.



Determine if the following functions fit the conditions of the Intermediate Value Theorem,

and determine the range of the results.

7. $f(x) = x^3 - 3x^2 + 2x + 5$ $\left| \quad x \in [0, 2] \right.$ 8. $f(x) = \sin(2\pi x)$ $\left| \quad x \in [-1, 1] \right.$
9. $g(t) = t\sqrt{t+6}$ $\left| \quad t \in [-4, 0] \right.$ 10. $H(t) = \frac{t}{(t-6)^2}$ $\left| \quad t \in [2, 8] \right.$

11. On May 15th, the weather in the town of Tarcía changes at a rate of $W(t)$ degrees Fahrenheit per hour. $W(t)$ is a twice-differentiable, increasing, and concave up function with selected values in the table below.

t (hours since midnight)	0	1	3	6	8
$W(t)$ (deg Fahrenheit per hour)	-2.4	-2.1	-1.2	1.8	4.5

At midnight ($t = 0$), the weather in Tarcía is 40 degrees Fahrenheit.

- (a) Is there a time in $0 \leq t \leq 8$ when the rate of change of the temperature equals 7 degrees Fahrenheit per hour? Justify your answer.
- (b) Is there a time in $0 \leq t \leq 8$ when $W(t) = 0$? Justify your answer.

12. Star Formation Rate (SFR) observations of red-shift allow scientists to track the total mass gained in a galaxy by the making of new stars. Below is a table of such data:

t	0	1	2	3	4	5	6	7	8
SFR	0.0029	0.0051	0.0055	0.0049	0.0042	0.0035	0.0029	0.0025	0.0021

SFR is measured in solar masses per cubic parsec per gigayear (millions of years), and t is measured in gigayears.

- (a) Is there a time in $0 \leq t \leq 8$ when $SFR = 0$? Justify your answer.
- (b) Is there a time in $0 \leq t \leq 8$ when $SFR = 0.0056$? Justify your answer.

4.2 Multiple Choice Homework

1. Let f be a polynomial function with degree greater than 2. If $a \neq b$ and $f(a) = f(b) = 1$, which of the following must be true of at least one value of x between a and b ?

- I. $f(x) = 0$ II. $f'(x) = 0$ III. $f''(x) = 0$

- a) I only b) II only c) III only d) II and III only e) I, II, and III
-

2. Which of the following functions **fail** to meet the conditions of the Mean Value Theorem?

I. $3x^{\frac{2}{3}} - 1$ $x \in [-1, 2]$

II. $|3x - 2|$ $x \in [1, 2]$.

III. $4x^3 - 2x + 3$ $x \in [-1, 2]$.

- a) I only b) II only c) III only d) I and II only e) II and III only
-

3. $y = -2x^2 + x - 2$ is defined on $x \in [1, 3]$. Find the c -value determined by the Mean Value Theorem.

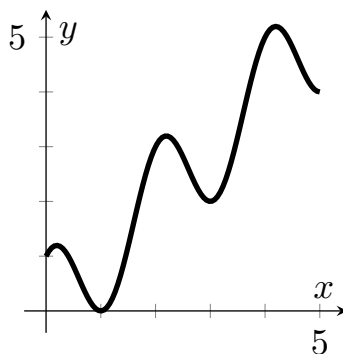
- a) $c = \frac{1}{2}$ b) $c = \frac{5}{4}$ c) $c = \frac{3}{2}$ d) $c = 2$ e) $c = \frac{9}{4}$
-

4. Let $g(x)$ be a continuous function on the interval $[0, 1]$ and let $g(1) = 0$ and $g(0) = 1$. Which of the following statements is not necessarily true?

- a) There exists a number c on $[0, 1]$ such that $g(c) \geq g(x)$ for all $x \in [0, 1]$.
 b) For all a and b in $[0, 1]$, if $a = b$, then $g(a) = g(b)$.
 c) There exists a number c on $[0, 1]$ such that $g(c) = \frac{1}{2}$.
 d) There exists a number c on $[0, 1]$ such that $g(c) = \frac{3}{2}$.
 e) For all c in $(0, 1)$, $\lim_{x \rightarrow c} g(x) = g(c)$.
-

5. The graph of a differentiable function f is shown below on the closed interval $[0, 5]$. How many values of x in the open interval $(0, 5)$ satisfy the conclusion of the Mean Value

Theorem for f on $[0, 5]$.



- a) 1 b) 2 c) 3 d) 4 e) 5

6. To which of the following phrases does the equation $Y = \frac{f(b) - f(a)}{b - a}$ pertain?

- a) The Mean Value Theorem
 b) The Average Value Theorem
 c) The Intermediate Value Theorem
 d) The Extreme Value Theorem
 e) Rolle's Theorem

7. The table below shows particular values of a differentiable function $E(t)$. At how many times on the interval $t \in [0, 12]$ does the Mean Value Theorem guarantee that $E'(t) = 0$?

t (hours)	0	2	4	6	8	10	12	14
$E(t)$	10	4	10	5	7	3	7	5

- a) 0 b) 1 c) 2 d) 3 e) 4